Relative error

$$= \frac{\Delta l mean}{\bar{l}} = \pm \frac{.02}{2.29} = \pm 8.7 \times 10^{-3}$$

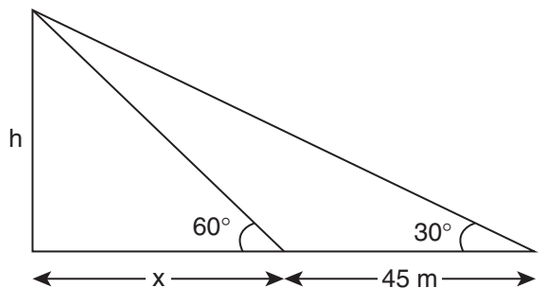
Percentage error $= \pm 8.7 \times 10^{-3} \times 100 =$

$0.87\% \times 100 = \pm(8.7 \times 10^{-1}) = 0.9\%$

7. The shadow of a pole standing on a level ground is found to be 45 m longer when the sun's altitude is 30° than when it was 60°. Determine the height of the pole. [Given $\sqrt{3} = 1.73$]

Solution:

Let the height of the pole be h.



Solution $\frac{x+45}{h} = \cot 30° \Rightarrow h = \frac{x+45}{\cot 30°}$

$\frac{x}{h} = \cot 60° \Rightarrow x = h \cot 60°$

Substituting the values of x in the above equation

$$h = \frac{h \cot 60° + 45}{\cot 30°}$$

$$h \cot 30° = h \cot 60° + 45$$

$$h(\cot 30° - \cot 60°) = 45$$

$$h = \frac{45}{\cot 30° - \cot 60°} = \frac{45}{\sqrt{3} - \frac{1}{\sqrt{3}}} = 38.97 \text{ m}$$

8. Calculate the number of times a human heart beats in the life of 100 years old man. Time of one heart beat = 0.8s.

Solution:

Life of the man = 100 years

100 years includes 76 normal years and 24 leap years

Total no of days $= 76 \times 365 + 24 \times 366$ = 36524 days

Number of seconds = 36524 × 24 × 3600 = 3.155 x 10^9 second

Number of heart beats =

$$= \frac{\textit{Total no of Seconds}}{\textit{Time period of heart beat}}$$

$$= \frac{3.155 \times 10^9}{0.8s} = 3.94 \times 10^9$$

9. The parallax of a heavenly body measured from two points diametrically opposite on equator of earth is 2′. Calculate the distance of the heavenly body. [Given radius of the earth = 6400km] [1″ = 4.85 x 10^{-6} rad]

Solution:

Angle $\theta = 2^1 = 2$ x 60″ = 120″ = $120 \times 4.85 \times 10^{-6}$ rad

$\theta = 5.82 \times 10^{-4}$ rad;

The distance of heavenly body

$$D = \frac{d}{\theta} = \frac{12800 \times 10^3}{5.82 \times 10^{-4}}$$

$$D = 2.19 \times 10^{10} \text{ m}$$

Appendix-1.indd 268 28-01-2020 14:13:29